



NAME:

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<<School Name>>

Level 1 Mathematics and Statistics

Practice Assessment 2023

91947

Demonstrate mathematical reasoning

5 credits

Credits: Five

Achievement	Achievement with Merit	Achievement with Excellence
Interpret and apply mathematical and statistical information in context.	Apply relational thinking to mathematical and statistical information in context.	Apply extended abstract thinking to mathematical and statistical information in context.

You should attempt ALL the questions in this booklet.

Make sure you have Resource Booklet 91947R.

If you need more room for any answer, use the extra space provided at the back of this booklet.

Check that this booklet has pages 2 to 7 in the correct order and that none of these pages is blank.

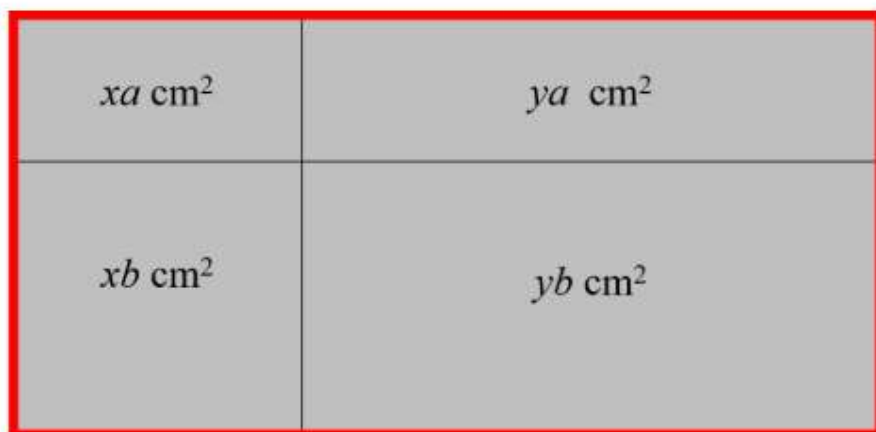
YOU MUST HAND THIS BOOKLET TO THE SUPERVISOR AT THE END OF THE EXAMINATION.

Links made from Subject Learning Outcome for Assessments document

<https://ncea-live-3-storagestack-53q-assetstorages3bucket-2o21xte0r81u.s3.amazonaws.com/s3fs-public/2024-04/MS%20NCEA%20Level%201%20Subject%20Learning%20Outcomes.pdf?VersionId=QXvuELZpwFBw66s9JLnV2XFqE4crufa2>

QUESTION 1

- (a) Consider the rectangle outlined in thick red, shown below. It is made up of four rectangles with areas of $xa \text{ cm}^2$, $ya \text{ cm}^2$, $xb \text{ cm}^2$, $yb \text{ cm}^2$. Write a simplified expression for the perimeter.



- rearranging, and using formulas (excluding the use of logarithms) – examples include: $d = \frac{1}{2}at^2$, $A = P(1 + nr)$, $t = 2\pi\sqrt{\frac{L}{g}}$
- expanding and simplifying expressions up to two brackets – examples include but not restricted to: $(2x^2 - 1)(x + 7)$, and $4x(x - 2) - (x - 2)$

Perimeter =

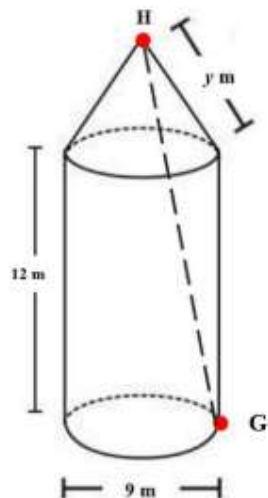
- (b) (i) Three positive numbers $\{a, b, c\}$ are said to be a Pythagorean triple if they follow the relationship $a^2 + b^2 = c^2$.

Find b if $\{20, b, 29\}$ is a Pythagorean triple.

- use Pythagoras' theorem in right-angled triangles in 2D and 3D situations

- (b) (ii) A cone is stacked on top of a cylinder as shown in the diagram below. The slant height of the cone placed on top is y metres, and the height of the cylinder is 12 metres. Show that the distance HG is

given by the expression $\sqrt{y^2 + 24\sqrt{y^2 - 20.25} + 144}$

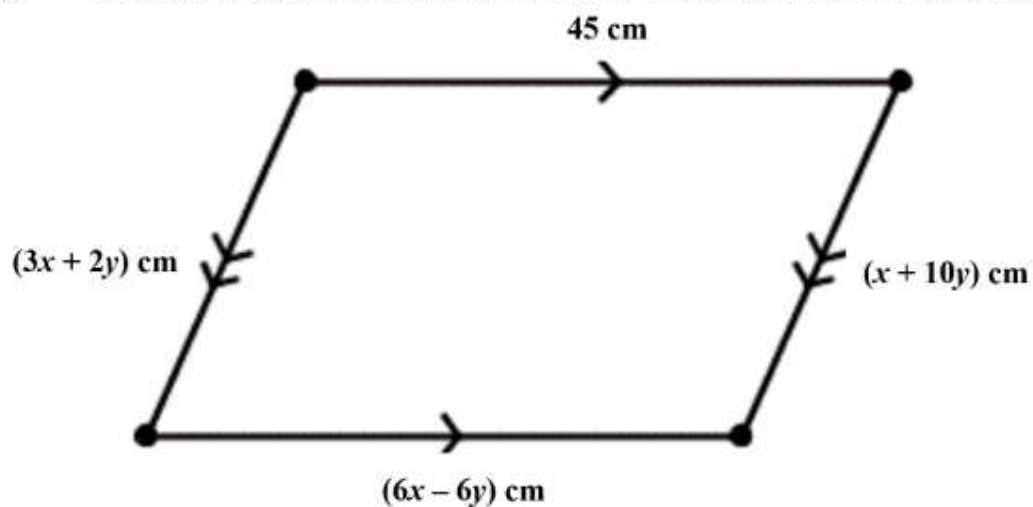


- use Pythagoras' theorem in right-angled triangles in 2D and 3D situations

- rearranging, and using formulas (excluding the use of logarithms) – examples include: $d = \frac{1}{2}at^2$, $A = P(1 + nr)$, $t =$

$$2\pi\sqrt{\frac{L}{g}}$$

- (c) For the parallelogram shown below, form equations and then find the values of x and y .



- use simultaneous linear equations with two unknowns, which includes solving given simultaneous equations, or forming and solving simultaneous equations. Methods for solving include substitution, elimination, graphical representation, graphing calculator and will not be prescribed. Note that solving linear with non-linear simultaneous equations are not included.

QUESTION 2

- (a) Write t in terms of r and s for the formula $r = \frac{s^3 t^2}{4}$

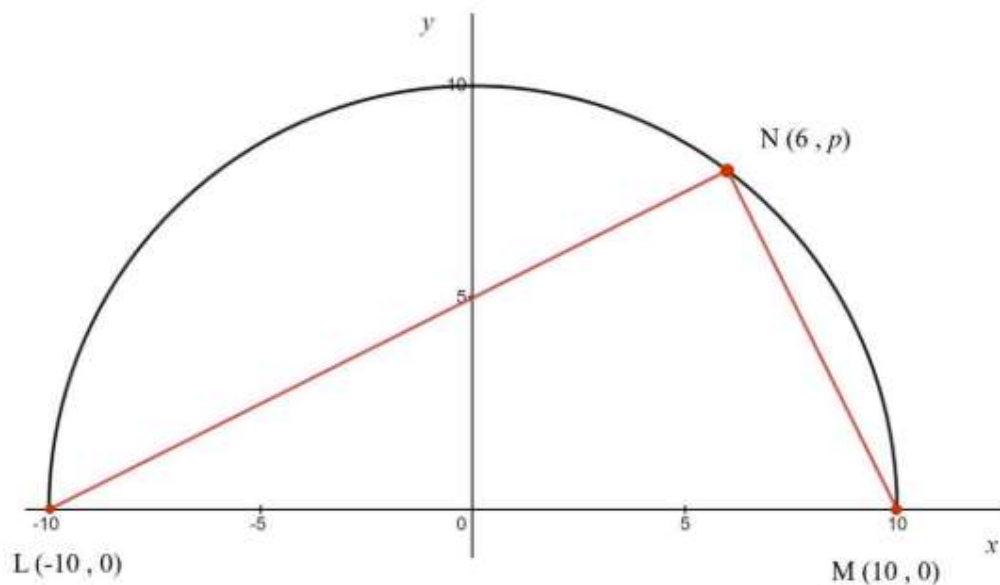
o rearranging, and using formulas (excluding the use of logarithms) – examples include: $d = \frac{1}{2}at^2$, $A = P(1 + nr)$, $t =$

$$2\pi\sqrt{\frac{L}{g}}$$

- (b) Three numbers have the property that when they are added in pairs, answers of 27, 35 and 42 are obtained. One statement from this information could be $a + b = 27$. What are the three numbers?

• use simultaneous linear equations with two unknowns, which includes solving given simultaneous equations, or forming and solving simultaneous equations. Methods for solving include substitution, elimination, graphical representation, graphing calculator and will not be prescribed. Note that solving linear with non-linear simultaneous equations are not included.

(c) Point N lies on a semi-circle as shown.



Show that triangle LMN is right angled.

- use Pythagoras' theorem in right-angled triangles in 2D and 3D situations

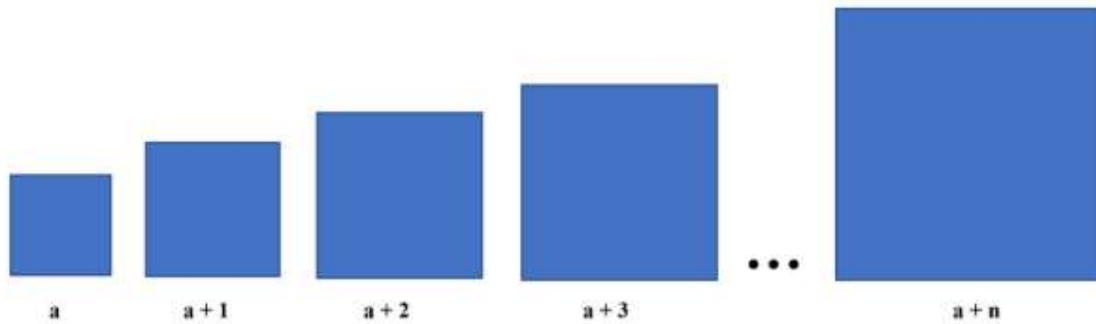
- expanding and simplifying expressions up to two brackets – examples include but not restricted to: $(2x^2 - 1)(x + 7)$, and $4x(x - 2) - (x - 2)$

(d) Solve the equation $\frac{a+5}{2} - \frac{9-a}{5} = \frac{3a+2}{20} + \frac{14}{5}$

- o operating on algebraic fractions with numeric denominators – examples include writing the following as a single fraction: $\frac{3x-2}{5} - \frac{x-1}{3}$ or $\frac{x^3}{3} + \frac{2x^2}{4}$

QUESTION 3

The diagram below shows a pattern of growing squares with their side lengths given for each square.



(a) (i) Write down a simplified expression for the total area of the **first three** squares.

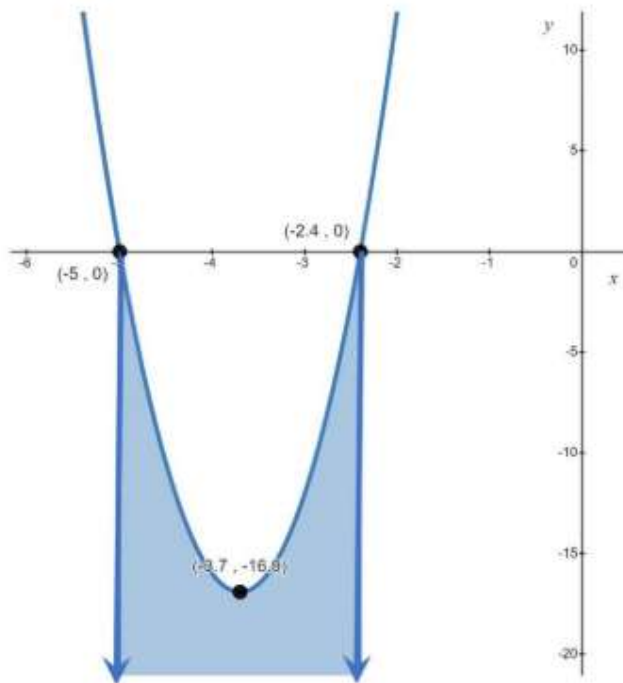
- expanding and simplifying expressions up to two brackets – examples include but not restricted to: $(2x^2 - 1)(x + 7)$, and $4x(x - 2) - (x - 2)$

(a) (iii) What is the difference in area between the largest square with side length $(a + n)$ and the second largest square in this pattern of squares, in terms of a and n ?

- expanding and simplifying expressions up to two brackets – examples include but not restricted to: $(2x^2 - 1)(x + 7)$, and $4x(x - 2) - (x - 2)$

(b) Write an inequality for the blue shaded region shown below underneath the parabola. Any equation for the parabola should be written in the form

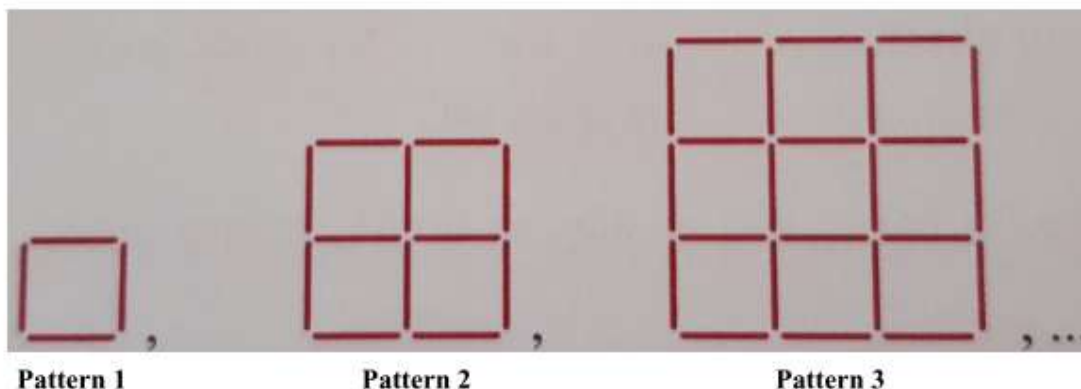
$$y = (ax + b)(cx + d) \quad \text{or} \quad y = ax^2 + bx + c \quad \text{or} \quad y = a(x - b)^2 - c$$



- o finding the equations for linear and quadratic functions, including horizontal and vertical lines, from patterns, tables or graphs

[illegible]

Consider the pattern made up of matchsticks shown below.



(c) (i) Give the formula for the n^{th} term for this pattern of matchsticks.

- o finding the equations for linear and quadratic functions, including horizontal and vertical lines, from patterns, tables or graphs

(c) (ii) Determine what pattern number contains 480 matchsticks

- use linear and quadratic equations, which includes solving linear or quadratic equations, or forming and solving linear and quadratic equations from statements or drawings. Quadratic equations will have either $a = 1$ or be written in such a way that factorising is accessible for students without a graphing calculator.

Assessment Schedule – 2023

Mathematics and Statistics: Interpret and apply mathematical and statistical information in context (91947)

Evidence Statement

QUESTION 1	Expected Coverage	Achievement (u)	Merit (r)	Excellence (t)
(a)	Longer side = $x + y$ Shorter side = $a + b$ Perimeter = $2(x + y) + 2(a + b)$ [or correct equivalent]	At least one side of bigger rectangle correctly 'factorised'	Correct and simplified perimeter in factorised form. NB Do not accept for 'r' grade $P = 2x + 2y + 2a + 2b$	
(b) (i)	$29^2 = 20^2 + b^2$ $b^2 = 29^2 - 20^2$ $b = 21$	At least until correct (or equivalent of) $b^2 = 29^2 - 20^2$	Two or more distinct features (at least one similarity and at least one difference) between the search terms referenced and discussed. Do not accept $b = \pm 21$ as questions states three positive numbers.	
(b) (ii)	$y^2 = x^2 + 4.5^2$ $x = \sqrt{y^2 - 20.25}$ [1] $HG^2 = (12 + x)^2 + 4.5^2$ $= 144 + 24x + x^2 + 4.5^2$ [2] $= 164.25 + 24(\sqrt{y^2 - 20.25}) + (y^2 - 20.25)$ So $HG = \sqrt{144 + 24\sqrt{y^2 - 20.25} + y^2}$ as required.	Line [1] correct awarded a 'u'	Line [2] correct argumentation awarded 'r'	Full and correct argument to required statement.

(c)	$3x + 2y = x + 10y \quad (1)$ $6x - 6y = 45 \quad (2)$ Rearrange (1) to $2x - 8y = 0$ So $x = 4y$ (3) Substitute into (2) Hence $24y - 6y = 45$ $y = 2.5$ and so $x = 10$	Setting up both equations (1) and (2)	Some correct algebraic manipulation to get to stage of solving for x and y . For example (3).	Correct values for x and y
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QUESTION 2	Expected Coverage	Achievement (u)	Merit (r)	Excellence (t)
(a)	$s^3 t^2 = 4r$ $t^2 = \frac{4r}{s^3}$ $t = \pm \sqrt{\frac{4r}{s^3}}$	Correct rearrangement for t (no need for $\pm$)		
(b)	$a + b = 27$ $a + c = 35$ $b + c = 42$ So $c = 42 - b$ $a + 42 - b = 35$ $a + b = 27$ So $2a = 20$, $a = 10$ Then $b = 17$ and $c = 15$	Guess and check numerical approach only gives 'u' grade only.	Setting up and solving of equations for a, b, and c.	
(c)	Distance LM = 20 Distance LN = $\sqrt{p^2 + 256}$ Distance MN = $\sqrt{p^2 + 16}$ $\left(\sqrt{p^2 + 256}\right)^2 + \left(\sqrt{p^2 + 16}\right)^2 = 20^2$ $p^2 + 256 + p^2 + 16 = 400$ $2p^2 + 272 = 400$ $p^2 = 64$ $p = 8$ Substitute back in for LN and MN gives a right angled triangle according to Pythagoras. NB Do not accept any response for u grade (Or higher) that merely states triangles in a semi circle must have a right angle at the circumference, unless accompanied by a geometric proof. We would expect to see a Pythagoras approach to solve this problem.	Correct Pythagoras statement set up	Value for p obtained.	Correct value for p and check that fits with theorem of Pythagoras.

(d)	$\frac{5a + 25 - 18 + 2a}{10} = \frac{15a + 10 + 280}{100}$ $\frac{7a + 7}{10} = \frac{15a + 290}{100}$ $7a + 7 = \frac{15a + 290}{10}$ $70a + 70 = 15a + 290$ $55a = 220$ $a = 4$	Common denominator or equivalent on both sides forms	All denominators simplified and equation on one line only.	Correct value for a .
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QUESTION 3	Expected Coverage	Achievement (u)	Merit (r)	Excellence (t)
(a) (i)	$a^2 + (a + 1)^2 + (a + 2)^2$ $= a^2 + a^2 + 2a + 1 + a^2 + 4a + 4$ $= 3a^2 + 6a + 5$	Correct expression.		
(a) (ii)	$(a + n)^2 - (a + (n - 1))^2$ $2an + n^2 - 2an - 2a - (n - 1)^2$ $n^2 - 2a - (n^2 - 2n + 1)$ $= 2n - 2a - 1$	Top line expression formed and at least one part correctly expanded	Correct expression for difference found.	
(b)	$y \leq (2x + 10)(5x + 12)$ $y \leq 10x^2 + 74x + 120$ $y \leq 10(x + 3.7)^2 - 16.9$ Or equivalent such as 2 as a factor taken out of first example	Evidence towards forming equation for parabola such as $y = (x + 5)(x + 2.4)$	Correct equation but no inequality stated	Correct inequation stated
(d) (i)	$M = 2n^2 + 2n$ Symbol notation not important	$2n^2$ correctly stated with some evidence of reasoning	Correct nth term formula	
(d) (ii)	$480 = 2n^2 + 2n$ $n^2 + n - 240 = 0$ $(n - 15)(n + 16) = 0$ $n = 15, -16$ Reject $n = 16$ So 15 th pattern		At least to equivalent of 2 nd line.	Correct pattern number worked out with reasoning. Evidence of rejection of 16 as a solution.

For each question

N0 no relevant comments

N1 attempt at 1 question

N2 one of u

A3 two of u

A4 three of u

M5 one of r

M6 two of r

E7 [1t]

E8 [2t]

	Not Achieved	Achievement	Achievement with Merit	Achievement with Excellence
Score range	0 – 7	8 – 13	14 – 18	19-24